Lab Report 05

Course title: Computer Network Laboratory
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Submitted to-

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Title: Implementation of RIP (Routing Information Protocol)

Objective:

- To understand the working principle of RIP (Routing Information Protocol).
- To configure three routers and connect three separate LANs.
- To configure IP addresses on routers and PCs.
- To enable RIP routing on all three routers.
- To establish communication between PCs in different LANs.

Description:

RIP (Routing Information Protocol) is a dynamic routing protocol used to exchange routing information between routers. It uses hop count to select the best route. The maximum hop count is 15, while 16 means the destination is unreachable.

Three routers, three switches and six PCs were added to Cisco Packet Tracer. Two PCs were connected to each switch and each switch was connected to a router. The routers were connected to each other to form the network topology.

IP addresses, subnet masks and default gateways were configured for all PCs. The router interfaces were also assigned appropriate IP addresses. The network consists of three LAN networks (192.168.1.0, 192.168.2.0 and 192.168.3.0) and three router-to-router connection networks (10.0.0.0, 20.0.0.0 and 30.0.0.0). RIP was enabled on each router through the CLI using the following commands such as:

```
Router(config)# router rip
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
Router(config-router)# network 30.0.0.0
```

The appropriate network commands were configured on each router according to its connected networks. After configuring RIP, the routers exchanged routing information and learned about remote networks automatically.

Connection:

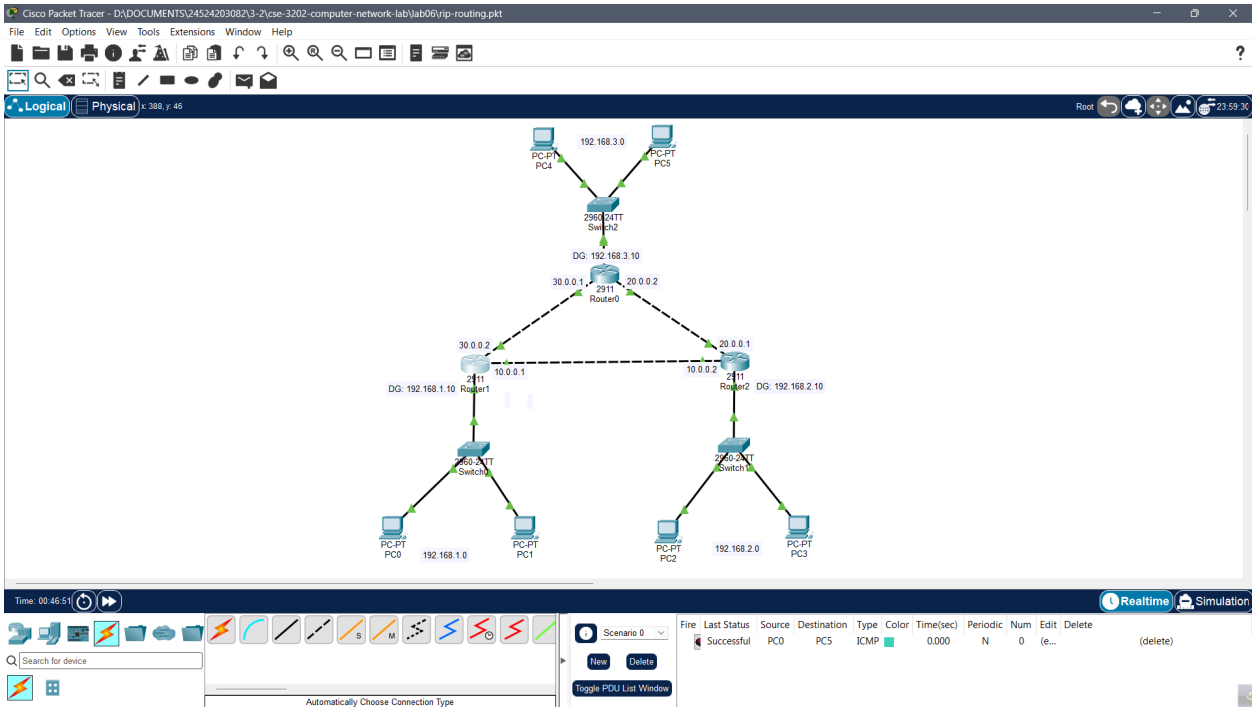


Figure 1: Complete Network Topology

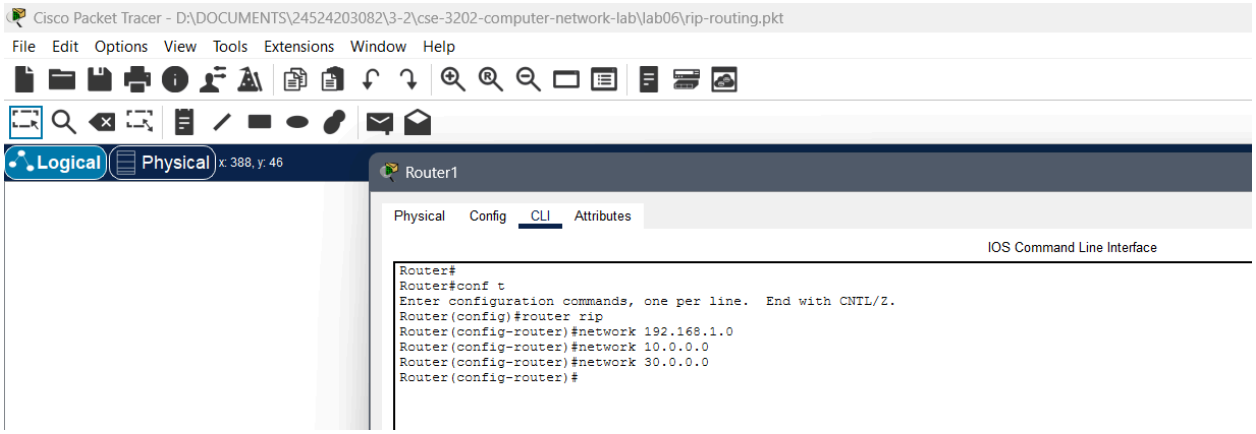


Figure 2: RIP Configuration on Routers

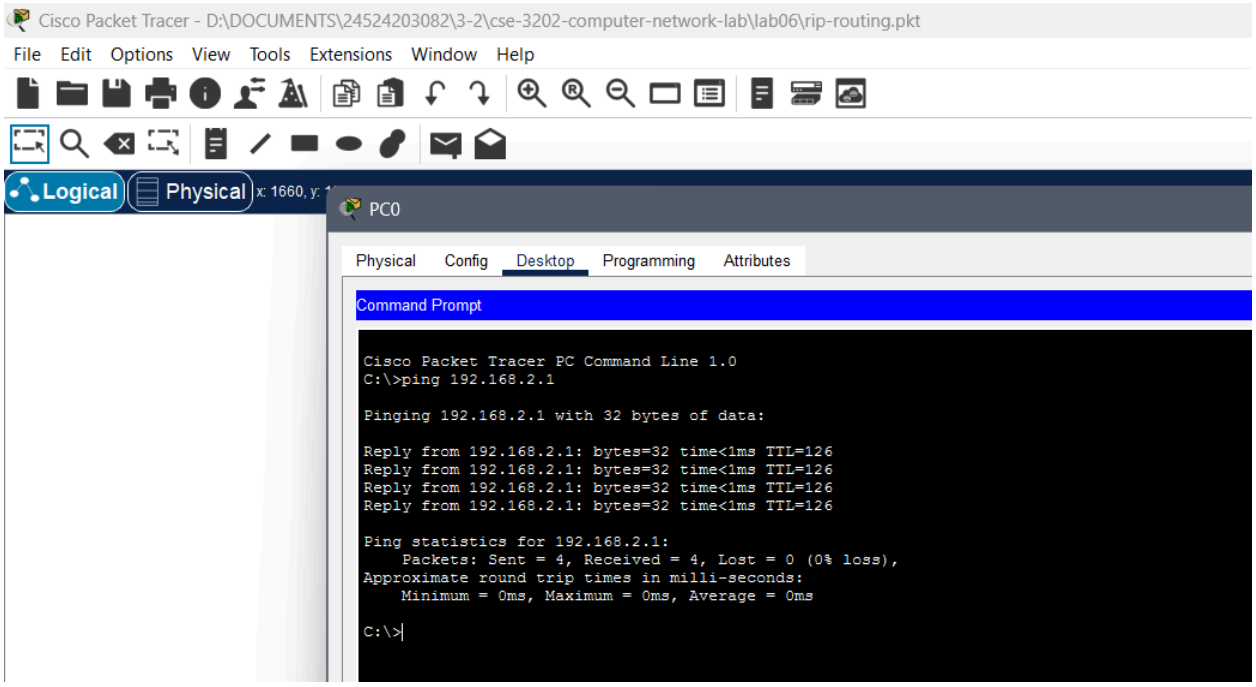


Figure 3: Ping Connectivity Test Between Different LANs

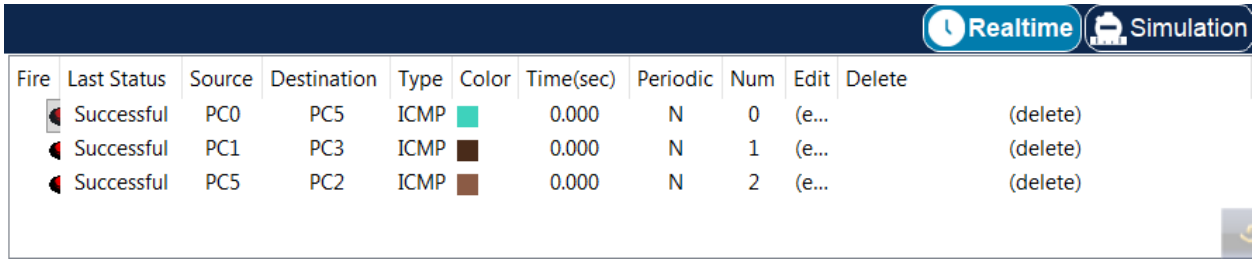


Figure 4: Successful Communication Between LANs